Lecture 25 (11/13)

Additional topics and practice

Today:

1) A practice problem on.
2) Double descent: what happens past the interpolation threshold?
3) Bayesian inference and the relationship to regularization.
4) The kernel trick

Question

Consider

$$Y = \beta_1 X + \epsilon, \quad \beta_0 = 0 \text{ and } E[X] = 0$$

so the true regression function is $f(x) = E[Y|X = x]$. Treat the X_i as fixed. Recall that the least squared estimate $\hat{\beta}_1$ has standard error $\text{se}(\hat{\beta}_1) = \sigma_\epsilon / \sqrt{N\hat{\sigma}_X^2}$.

(a) What are the bias and the variance of $\hat{\beta}_1$?

(b) Compute $\text{MSE}_{\hat{y}(3,D)} = E_D\left[(\hat{y}(3, D) - f(3))^2\right]$. What is $E\left[(\hat{y}(3, D) - Y)^2\right]$?

(c) What is the expected squared error $E\left[(\hat{y}(3, D) - Y)^2\right]$ for a new point at $x = 3$?

Solution

The prediction is $\hat{y}(x, D) = \hat{\beta}_1 x$.
(a) Least squares is unbiased, $E[\hat{\beta}_1] = \beta_1$, so the bias is 0. The variance is the square of the standard error:

$$\text{var}(\hat{\beta}_1) = \frac{\sigma_\epsilon^2}{N\hat{\sigma}_X^2}.$$

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(b) $\hat{y}(3, D) - f(3) = 3\hat{\beta}_1 - 3\beta_1 = 3(\hat{\beta}_1 - \beta_1)$, so

$$\text{MSE}_{\hat{y}(3,D)} = E_D\left[9(\hat{\beta}_1 - \beta_1)^2\right] = 9\,\text{var}(\hat{\beta}_1) = \frac{9\sigma_\epsilon^2}{N\hat{\sigma}_X^2}.$$

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(The bias is 0)
(c) A new point is $Y = f(3) + \epsilon$, with ϵ independent of the training set D, so the cross term has mean 0:

$$E\left[(\hat{y}(3, D) - Y)^2\right] = E_D\left[(\hat{y}(3, D) - f(3))^2\right] + E[\epsilon^2] = \frac{9\sigma_\epsilon^2}{N\hat{\sigma}_X^2} + \sigma_\epsilon^2.$$

PNAS

Reconciling modern machine-learning practice and the classical bias–variance trade-off

Mikhail Belkin, Daniel Hsu, Siyuan Ma, and Soumik Mandal Authors Info & Affiliations

Edited by Peter J. Bickel, University of California, Berkeley, CA, and approved July 2, 2019 (received for review February 21, 2019)

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PNAS

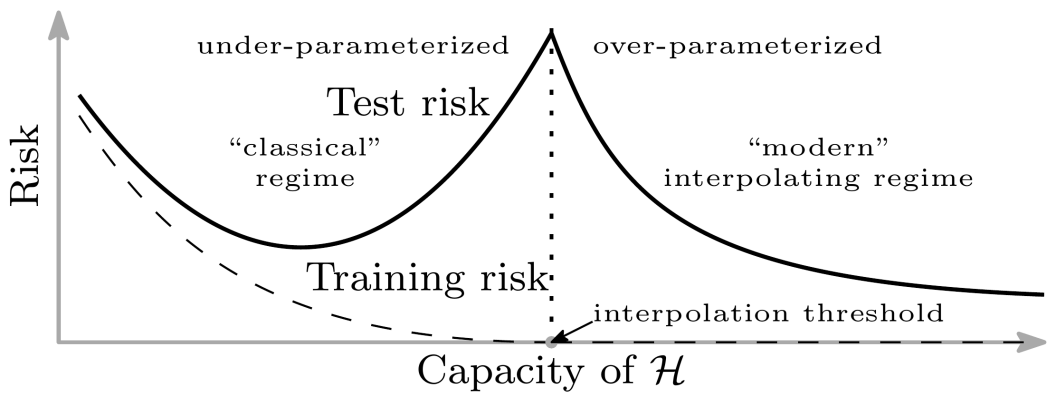
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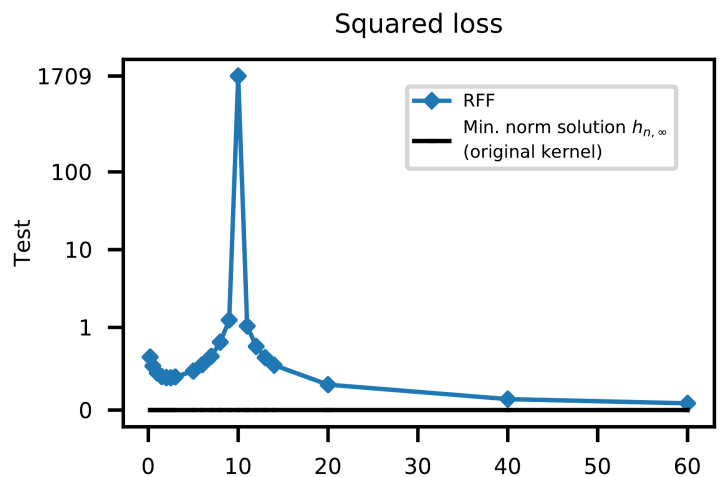
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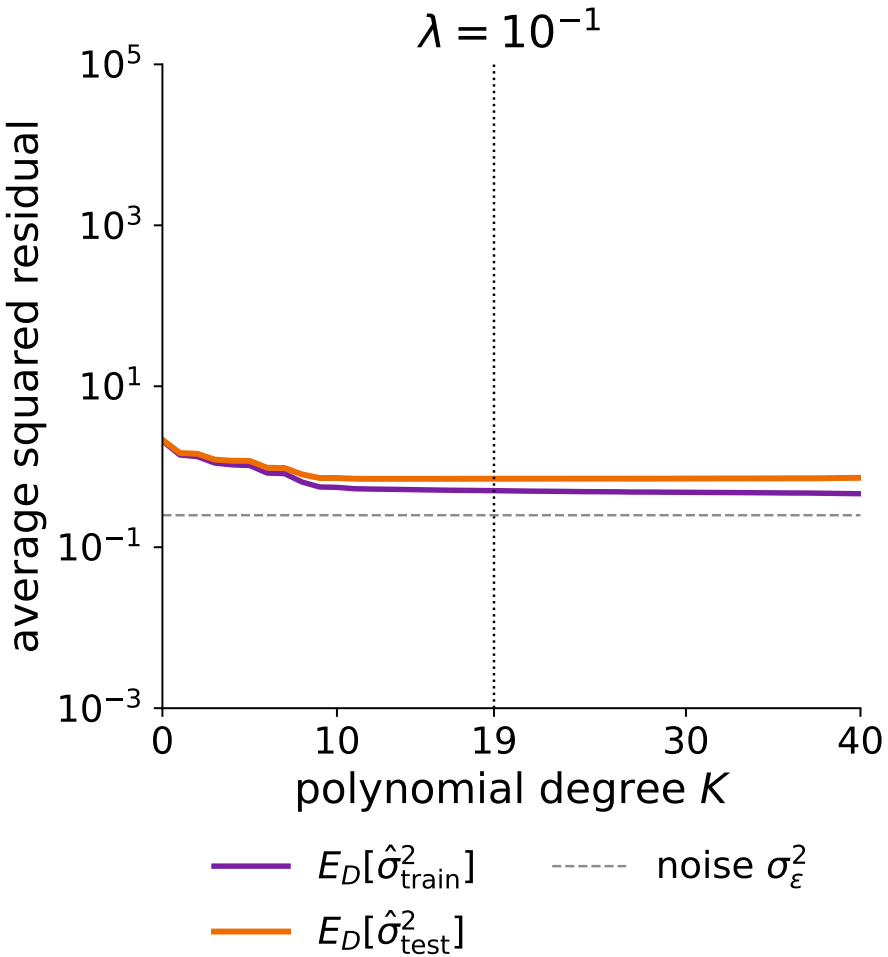
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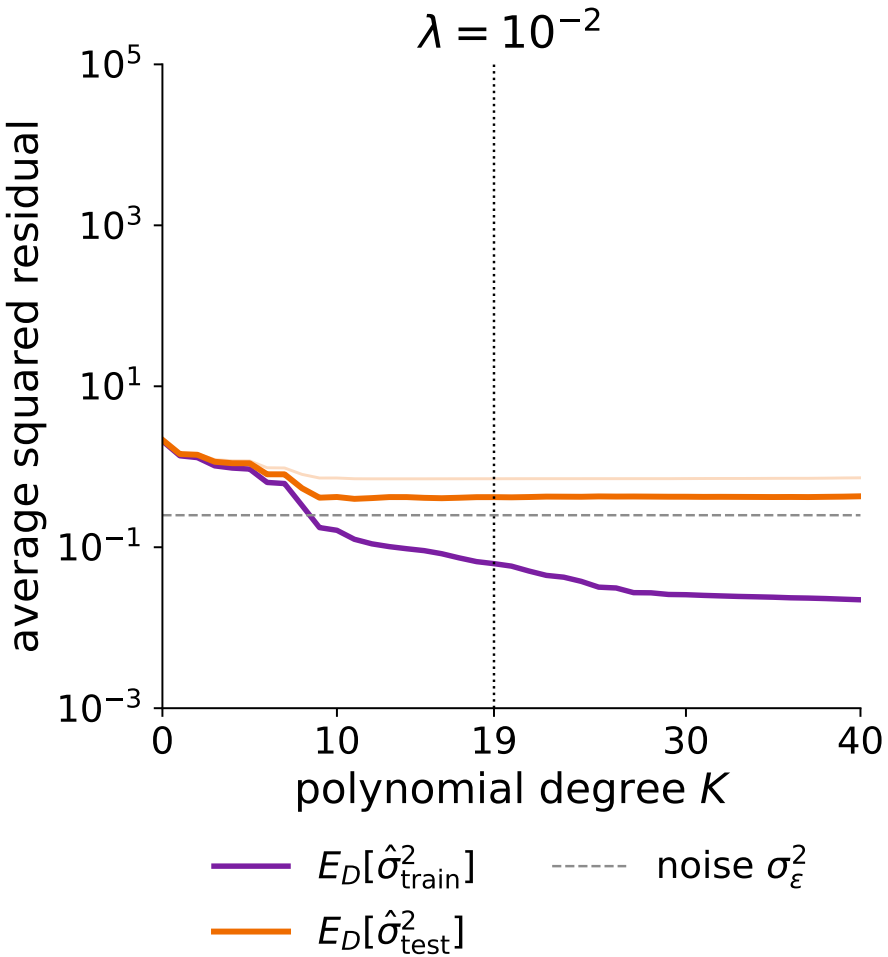
Double descent in polynomial regression

Consider polynomial regression with degree K.



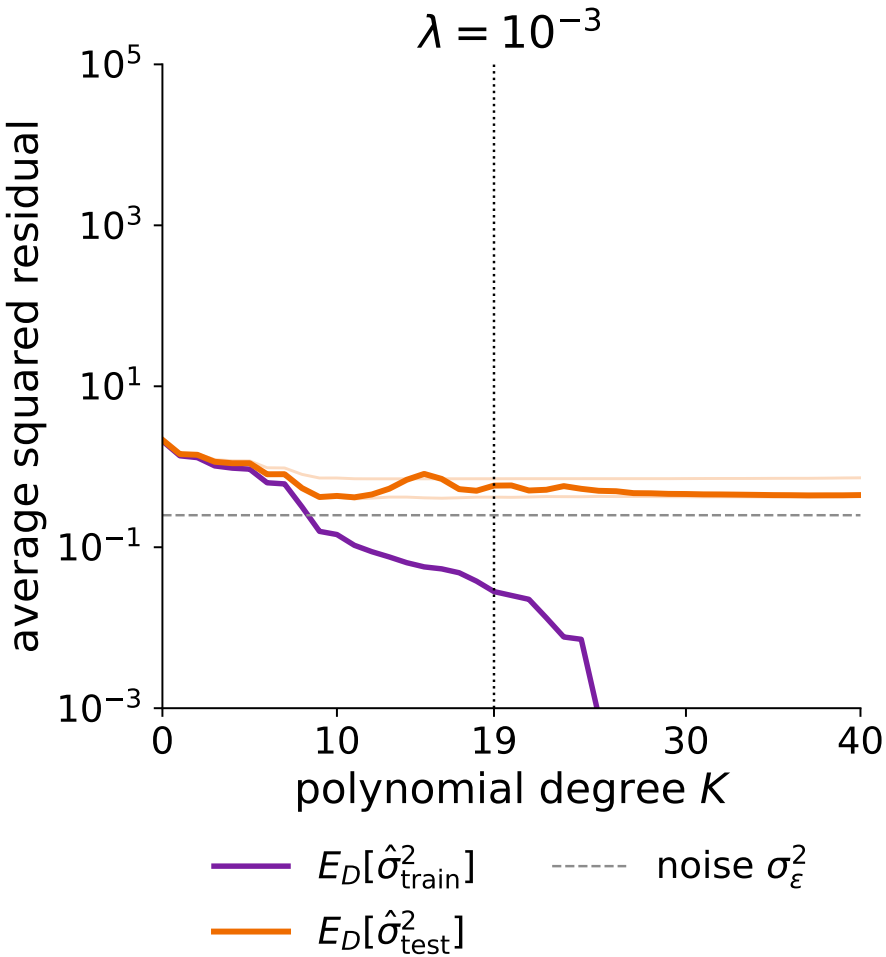
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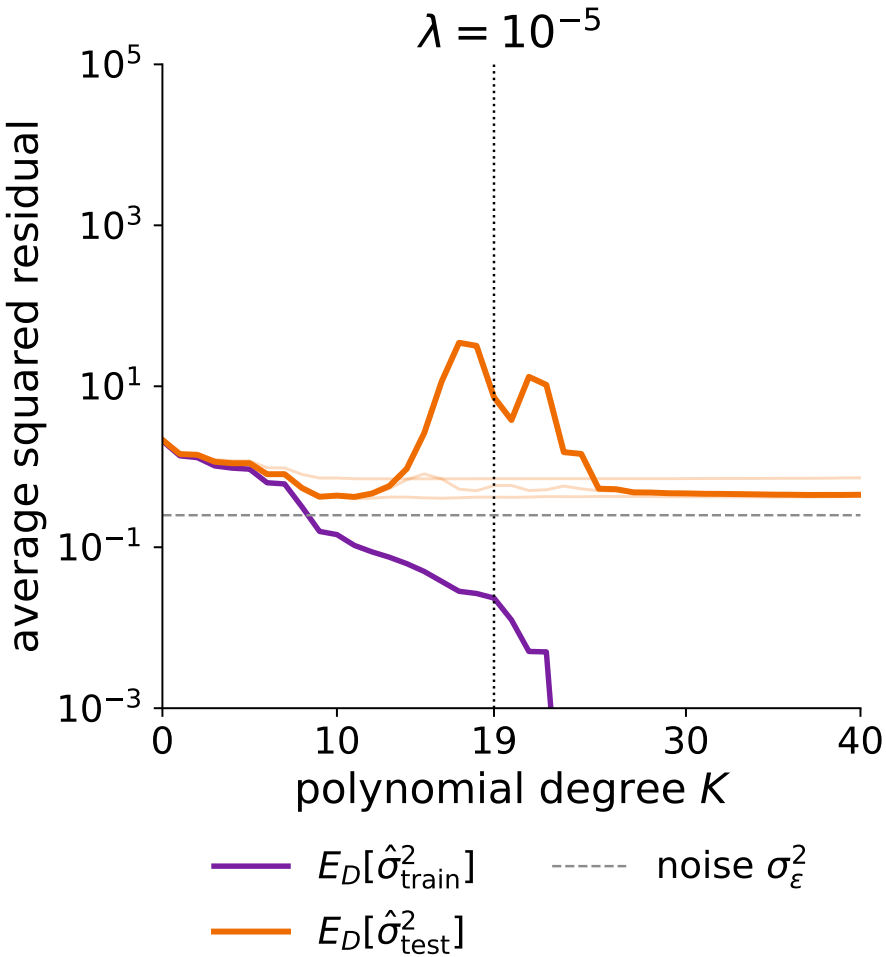
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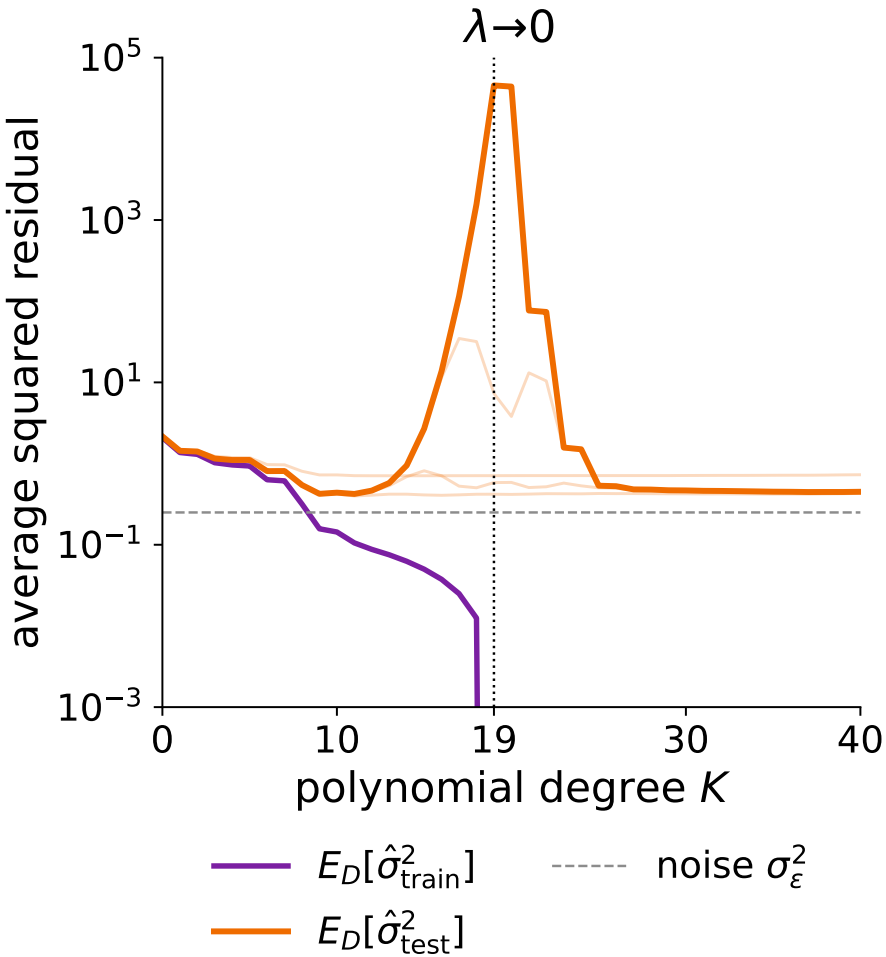
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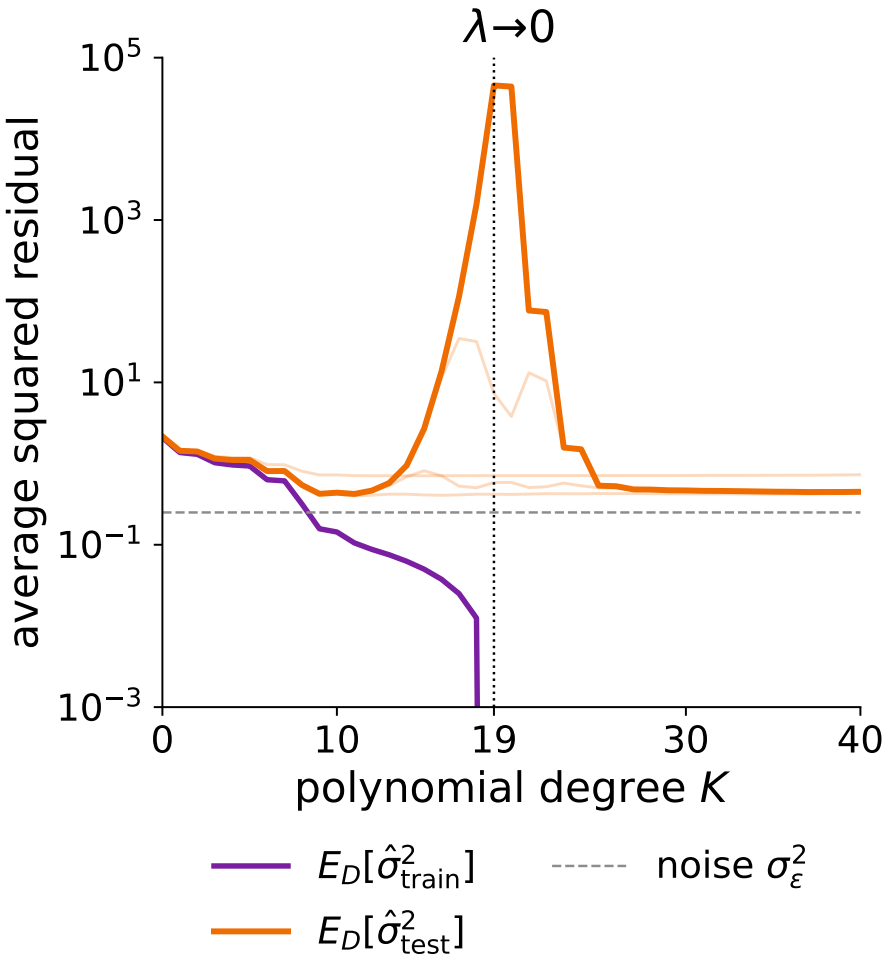
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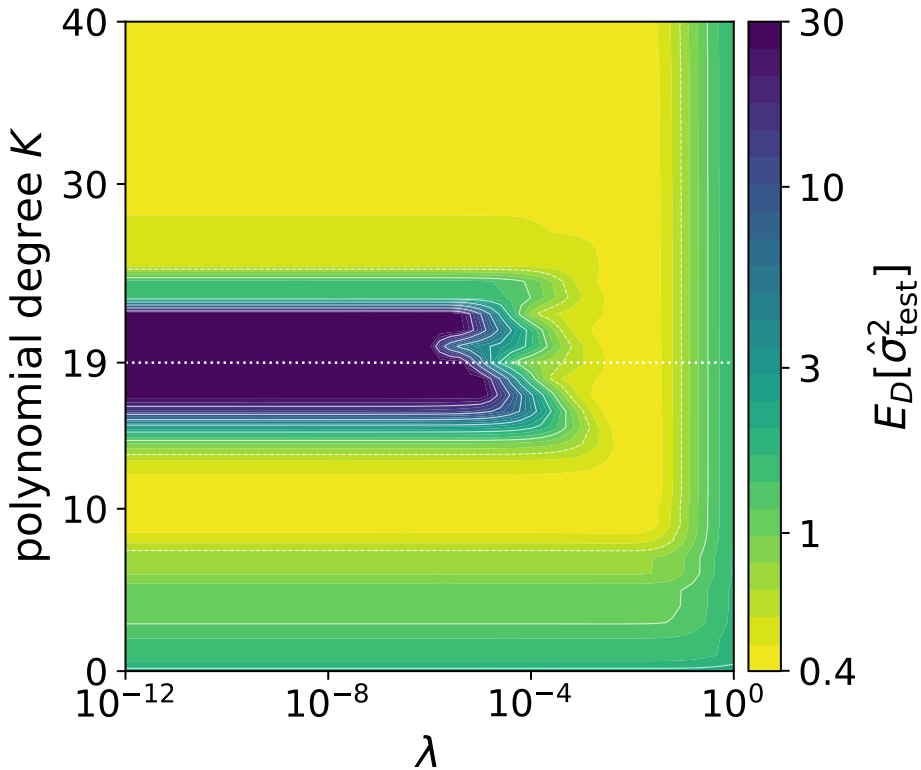
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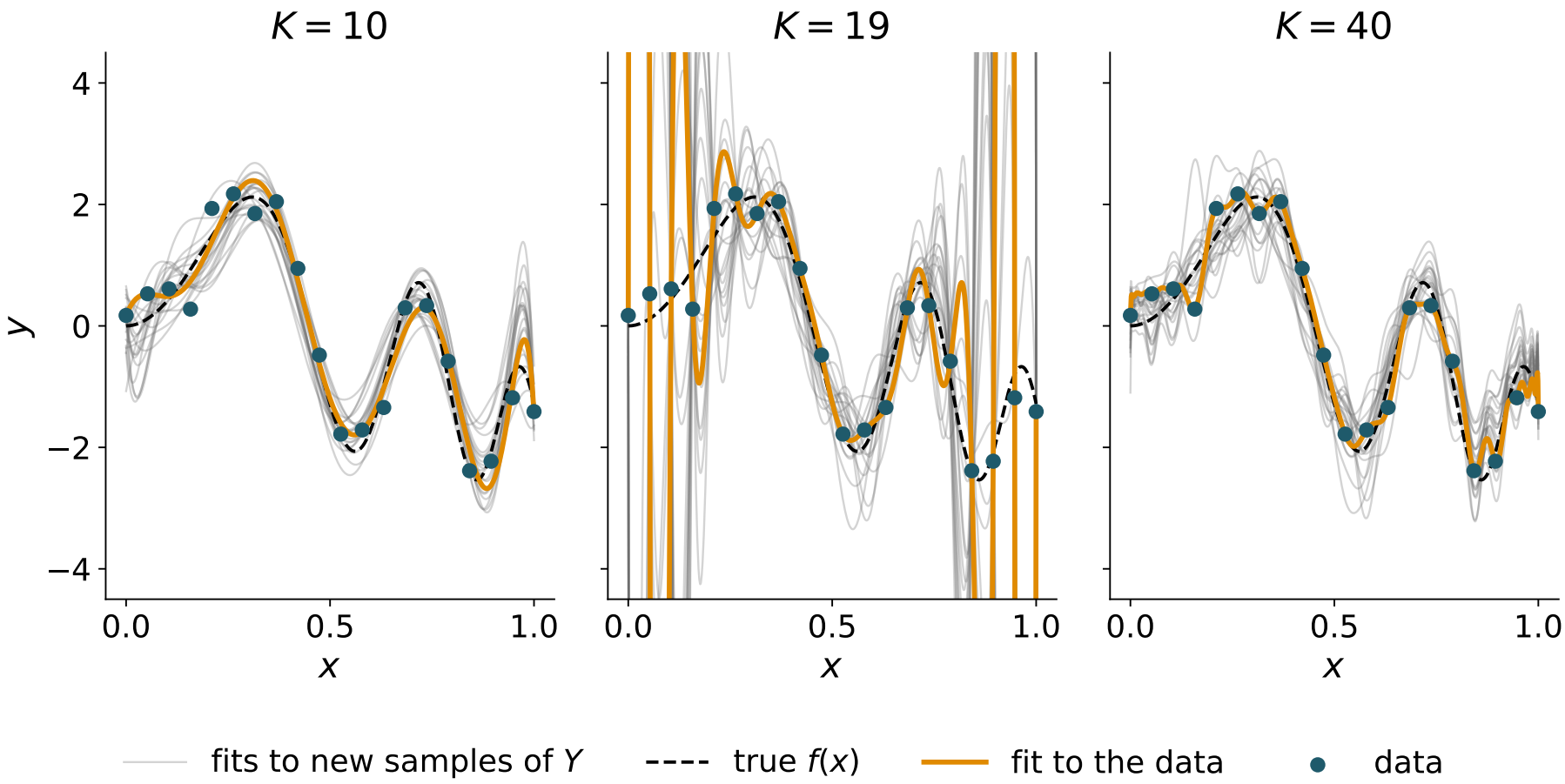


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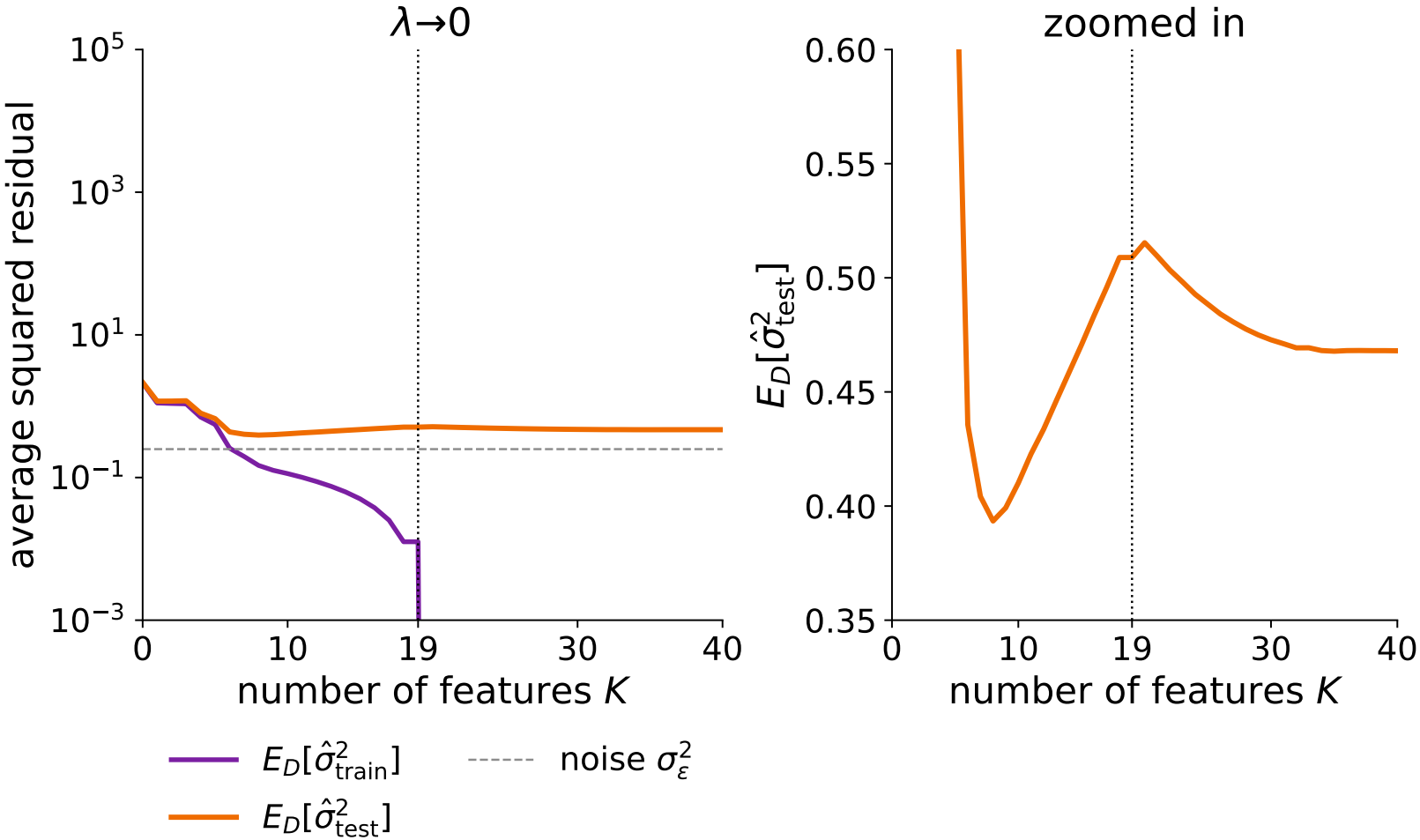




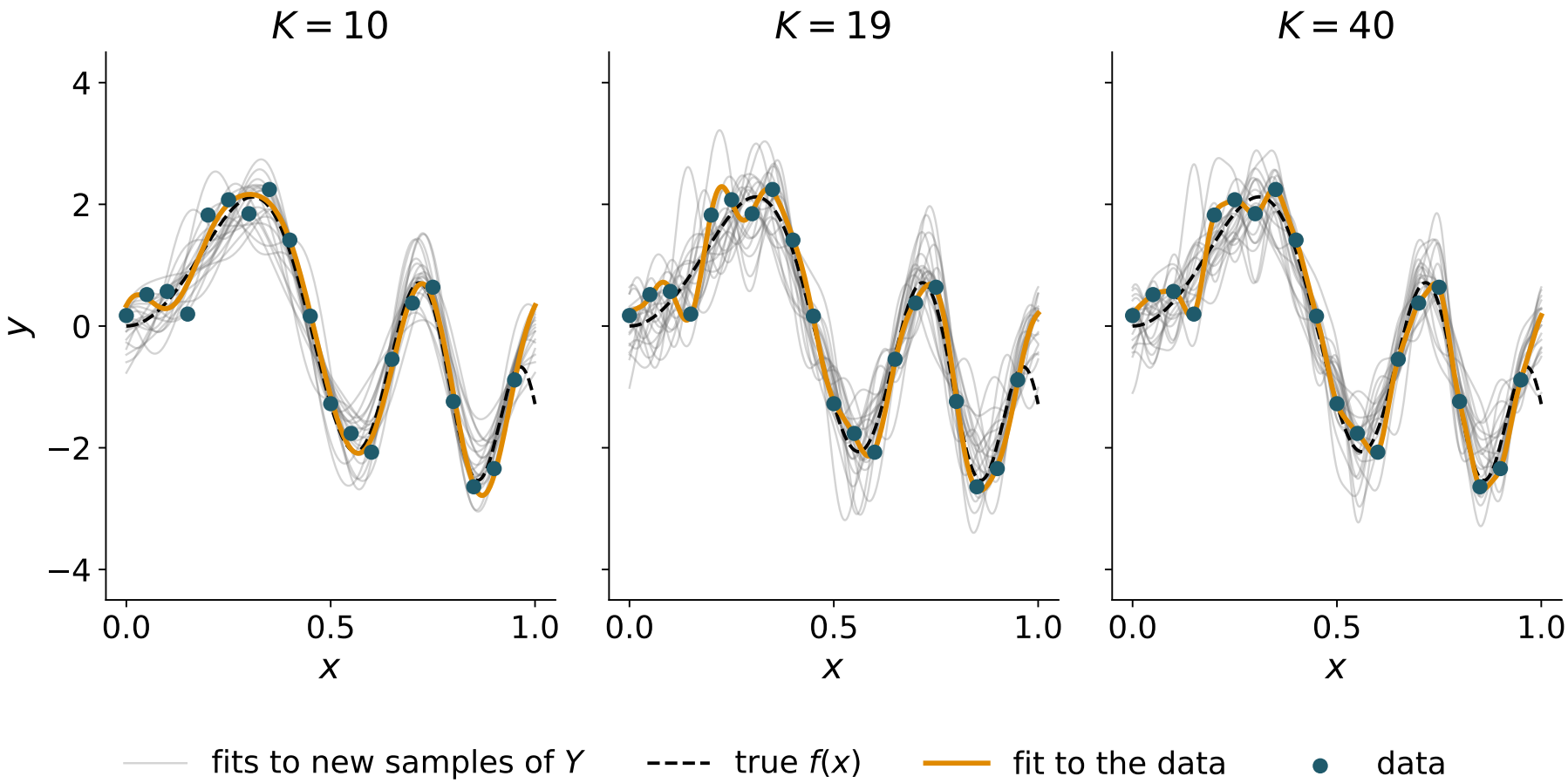

$K = 10$
$K = 19$
$K = 40$
y
x
4
2
0
−2
−4
0.0
0.5
1.0
fits to new samples of Y
true $f(x)$
fit to the data
data

Double descent in a Fourier (sin/cos) model

Consider a Fourier model with K features.



Double descent in a Fourier (sin/cos) model



Bayesian inference: The idea

- So far, we have thought of parameters as a fixed quantities which are estimated from data. The sample distribution provides the uncertainty in these estimates. **The estimator is thought of as a random variable, but NOT the parameter itself.**

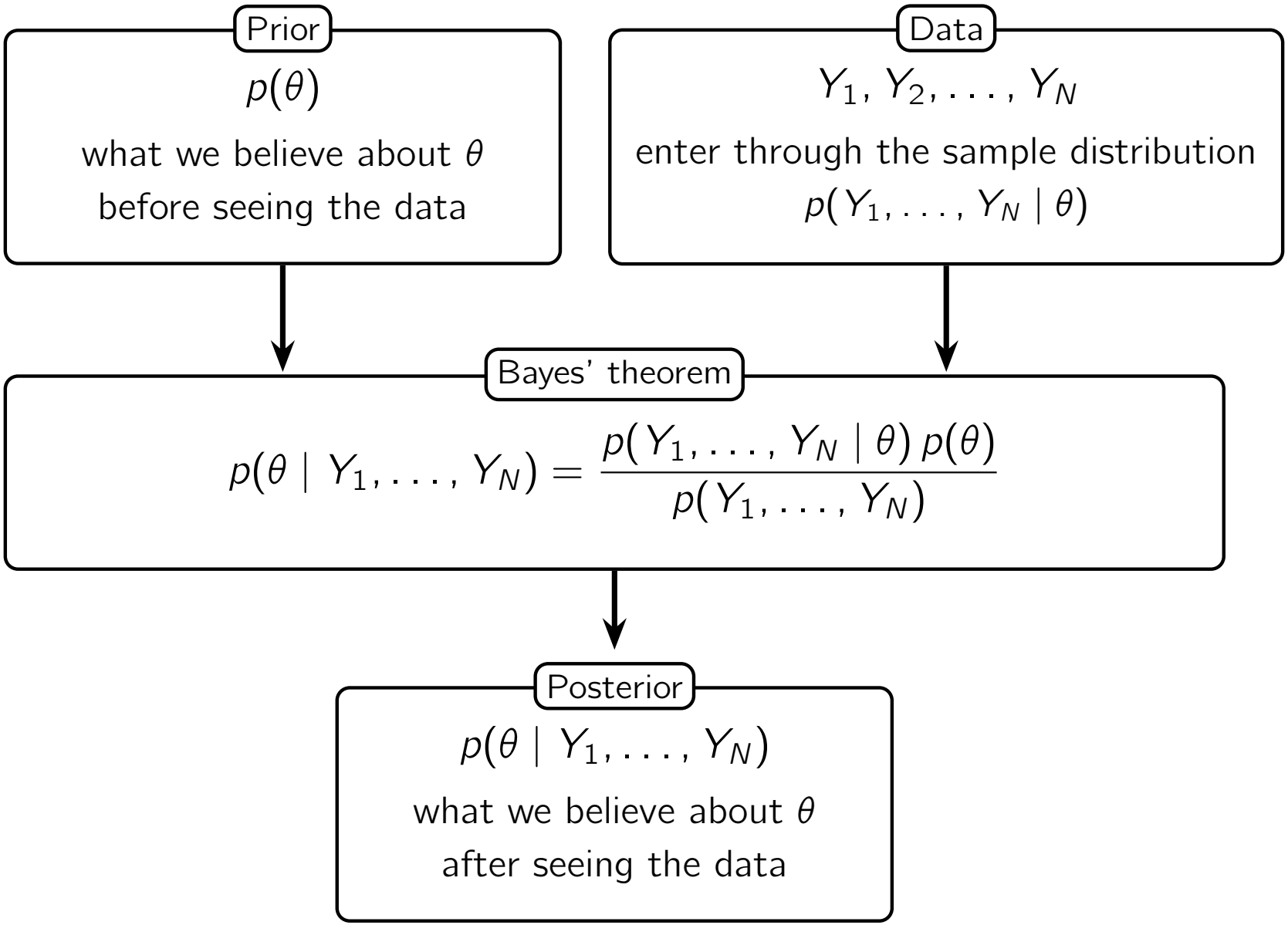
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- In Bayesian inference, the idea is to treat parameters themselves as random variables *before* fitting the data, and then update this distribution based on the data.
- The distribution of a parameter before we fit the data is a our prior.

Bayesian inference



Bayesian: Simple example

$$\mu \sim \text{Normal}(0, \tau^2) \quad \text{(prior)}$$

$$Y \mid \mu \sim \text{Normal}(\mu, \sigma_\epsilon^2) \quad \text{(likelihood)}$$

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Question

Assuming σ_ϵ^2 is fixed (meaning we don't model it with priors), what is the posterior of μ?

Finding the posterior for the mean

Both μ and Y have mean 0, and

$$\text{var}(Y) = \tau^2 + \sigma_\epsilon^2, \qquad \text{cov}(\mu, Y) = \text{cov}(\mu, \mu + \epsilon) = \tau^2.$$

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Regress μ (response) on Y (predictor). The slope is

$$\frac{\text{cov}(\mu, Y)}{\text{var}(Y)} = \frac{\tau^2}{\tau^2 + \sigma_\epsilon^2}, \qquad \text{and} \qquad \rho^2 = \frac{\text{cov}(\mu, Y)^2}{\text{var}(\mu)\text{var}(Y)} = \frac{\tau^2}{\tau^2 + \sigma_\epsilon^2}.$$

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The noise variance is $\text{var}(\mu)(1 - \rho^2)$, and (μ, Y) are jointly Normal, so

$$\mu \mid Y \sim \text{Normal}\Big(\frac{\tau^2}{\tau^2 + \sigma_\epsilon^2}\, Y,\ \frac{\tau^2 \sigma_\epsilon^2}{\tau^2 + \sigma_\epsilon^2}\Big).$$

Bayesian: Many observations

$$\mu \sim \text{Normal}(0, \tau^2) \qquad \text{(prior)}$$

$$Y_1, \ldots, Y_N \mid \mu \sim \text{Normal}(\mu, \sigma_\epsilon^2) \text{ independent} \qquad \text{(likelihood)}$$

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Question

What is the posterior of μ given $Y_1, \ldots, Y_N$?

Finding the posterior with many observations

The posterior depends on the data only through $\overline{Y}$, and $\overline{Y} \mid \mu \sim \text{Normal}(\mu, \sigma_\epsilon^2/N)$.

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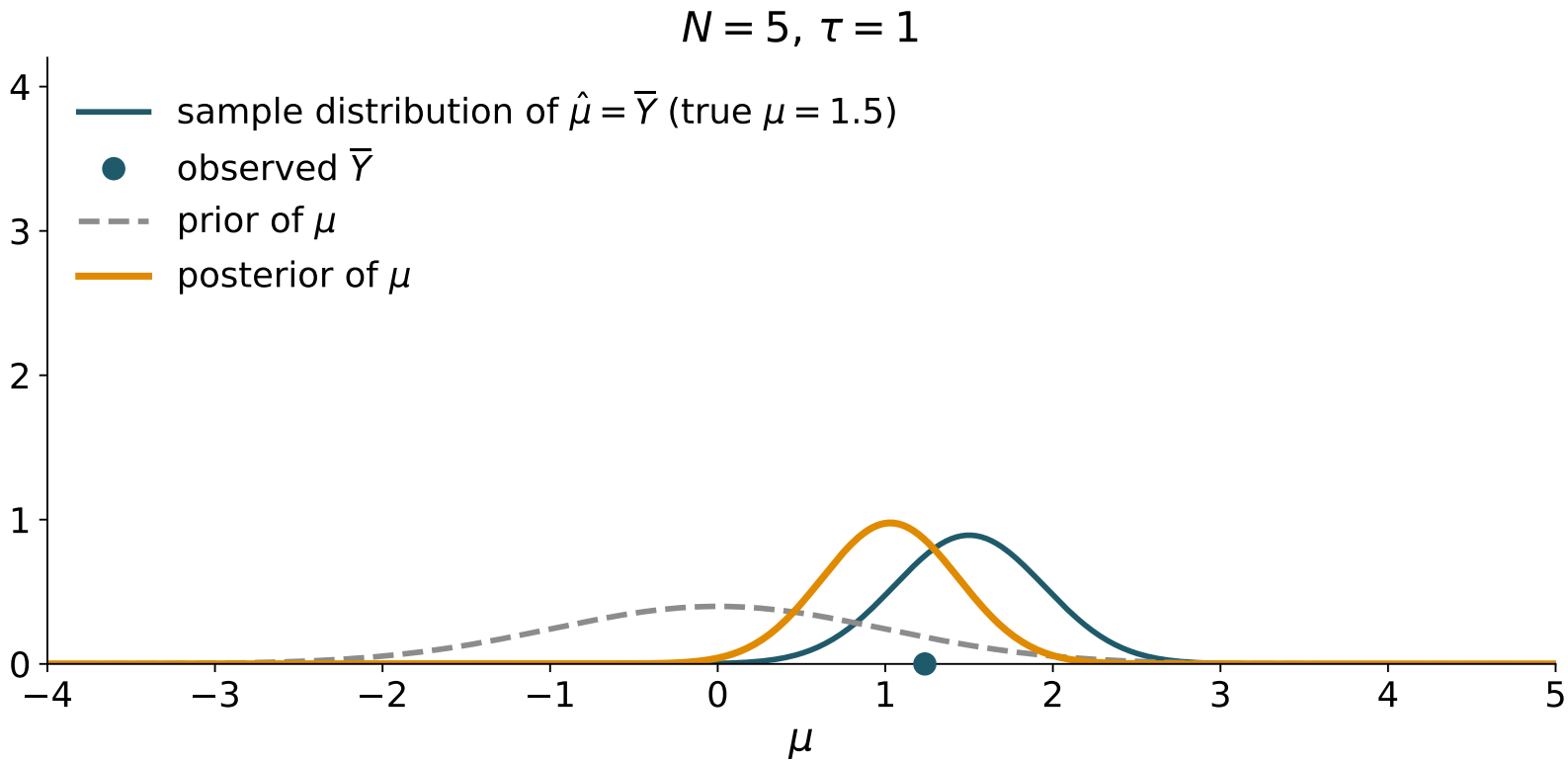
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Regressing μ (response) on $\overline{Y}$ (predictor) as before,

$$\mu \mid Y_1, \ldots, Y_N \sim \text{Normal}\Big(\frac{\tau^2}{\tau^2 + \sigma_\epsilon^2/N}\,\overline{Y},\ \frac{\tau^2 \sigma_\epsilon^2/N}{\tau^2 + \sigma_\epsilon^2/N}\Big).$$

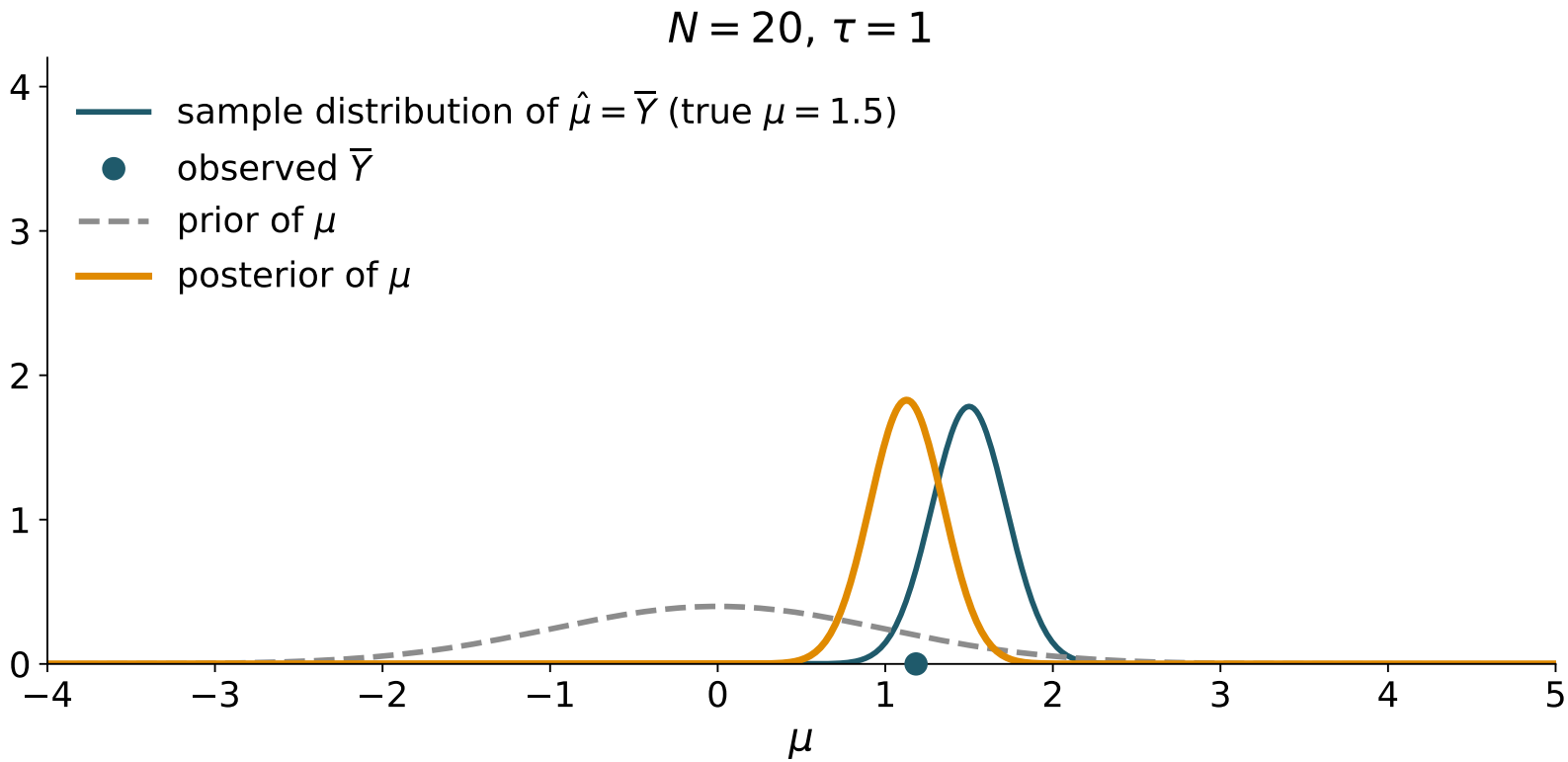
Prior, sample and posterior

$\sigma_\epsilon = 1.$



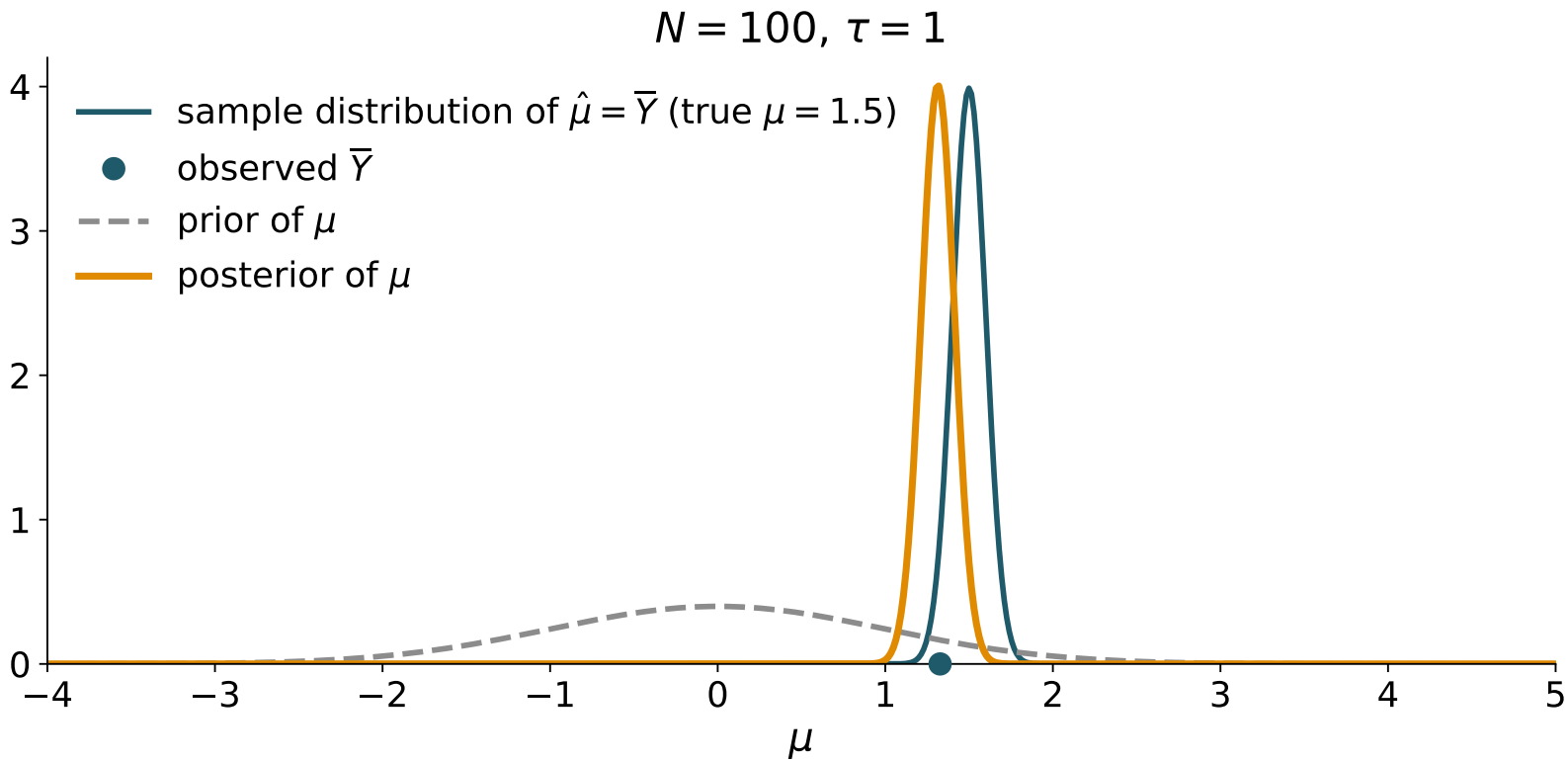
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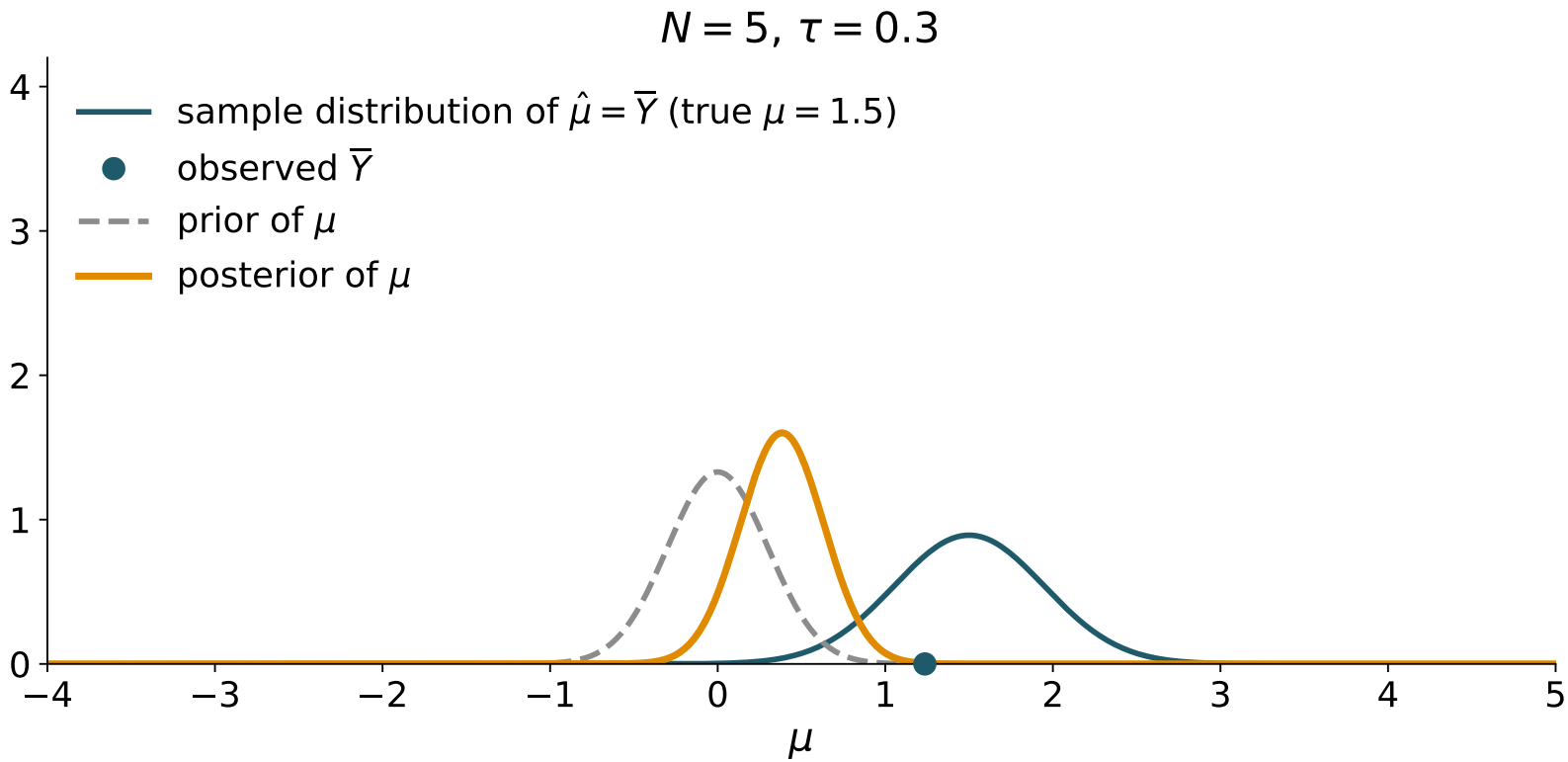
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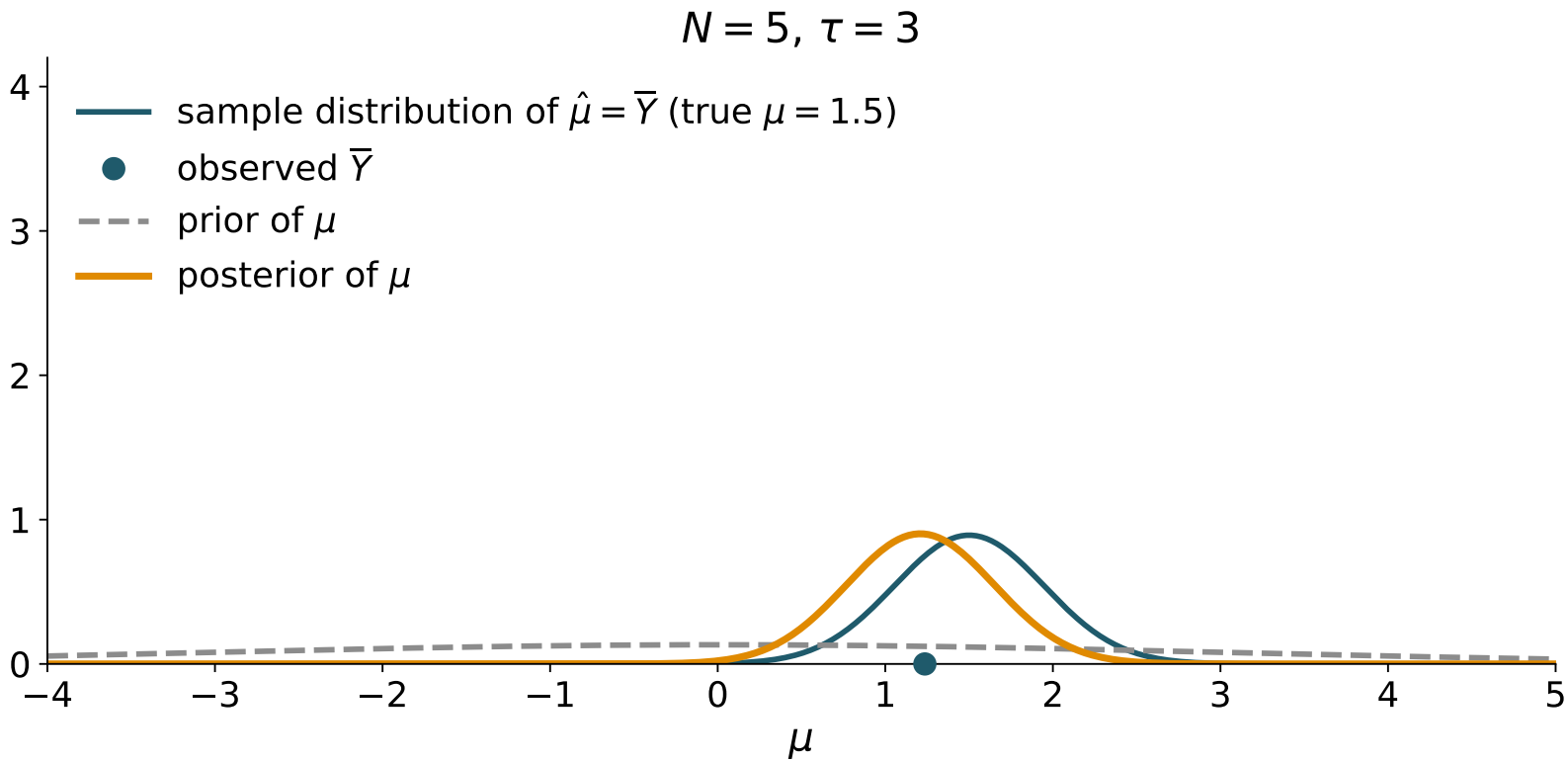
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Laplace's rule of succession from a prior

Take a uniform prior on q and N Bernoulli trials:

$$q \sim \text{Uniform}(0, 1)$$

$$Y_i \mid q \sim \text{Bernoulli}(q), \quad i = 1, \ldots, N, \qquad S = \sum_i Y_i$$

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$$E[q \mid S] = \frac{S + 1}{N + 2}.$$

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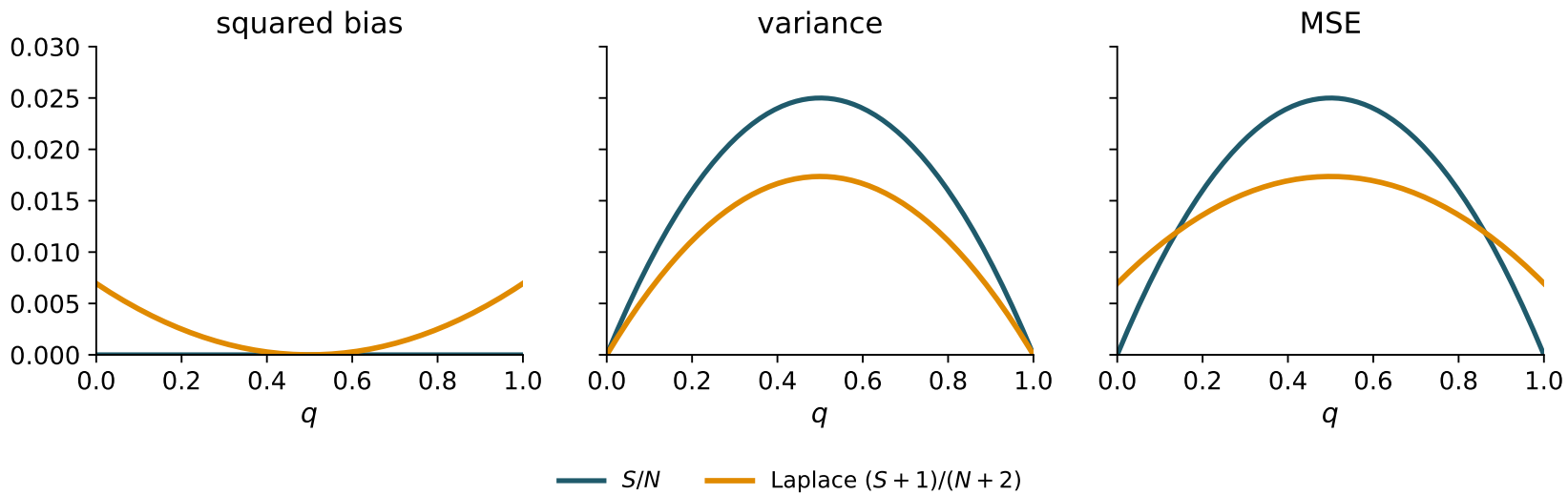
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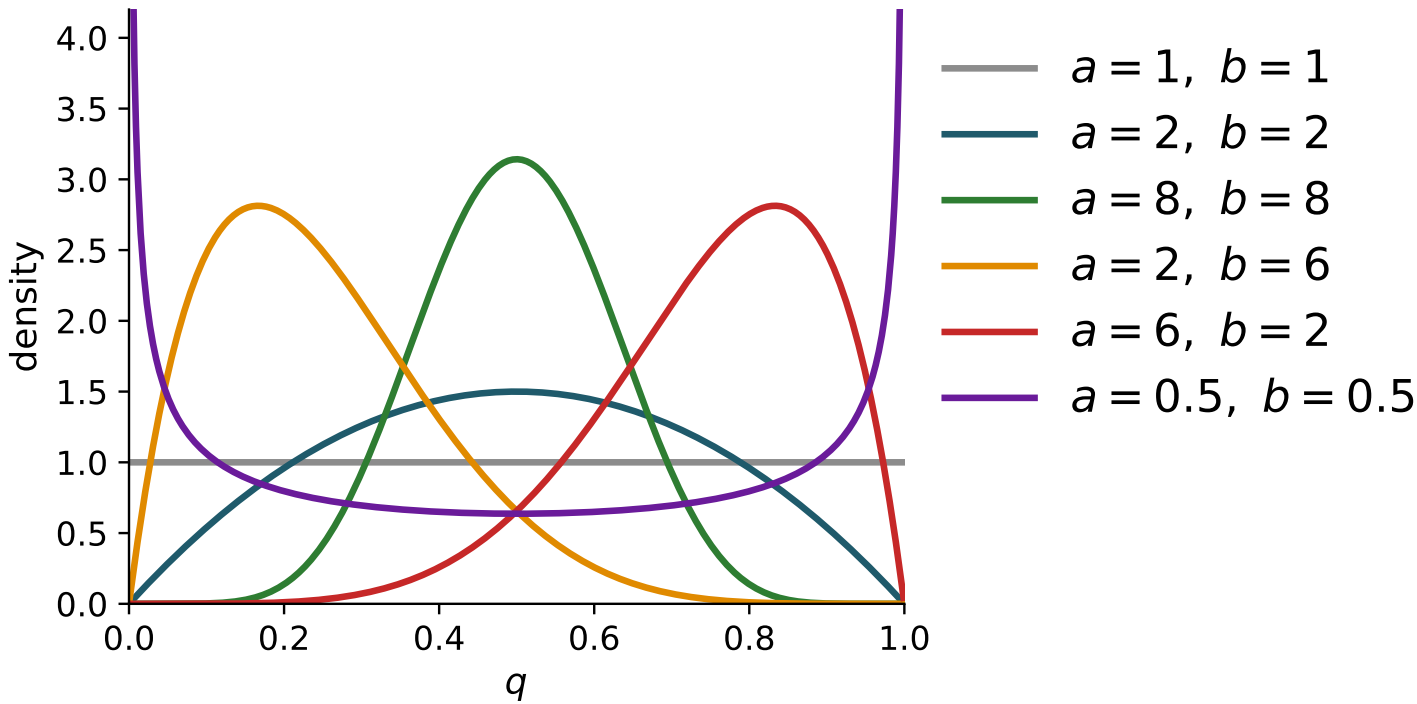


A more general prior: the Beta distribution

Replace the uniform prior by $q \sim \mathrm{Beta}(a, b)$, with density

$$f(q) \propto q^{a-1}(1-q)^{b-1}, \quad 0 < q < 1, \qquad E[q] = \frac{a}{a+b}.$$

$a = b = 1$ is the uniform prior. Think of a and b as prior successes and failures.



Posterior with a Beta prior

$$q \sim \text{Beta}(a, b)$$

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The posterior is $q \mid S \sim \mathrm{Beta}(a + S,\ b + N - S)$, so

$$E[q \mid S] = \frac{S + a}{N + a + b} = \frac{N}{N + a + b}\,\frac{S}{N} + \frac{a + b}{N + a + b}\,\frac{a}{a + b}.$$

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Posterior with a Beta prior

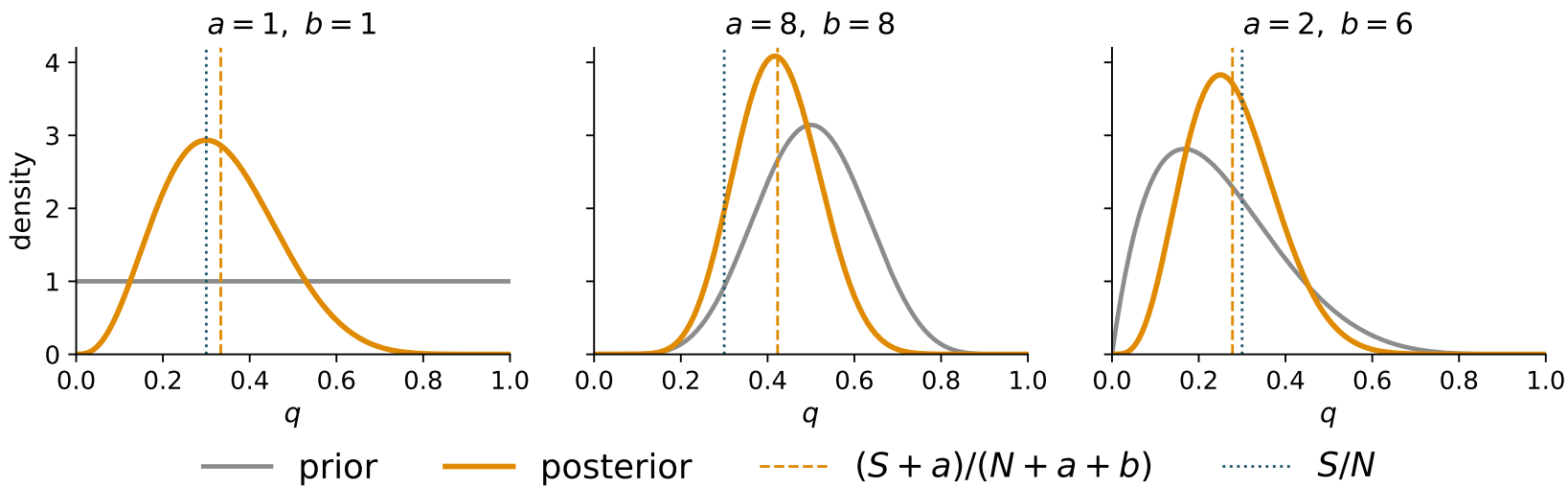
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Regularization from priors

Put a prior on each coefficient:

$$\beta_j \sim \text{Normal}(0, \tau_j^2), \quad j = 1, \ldots, K$$

$$Y \mid X, \beta \sim \text{Normal}\Big(\sum_j \beta_j \phi_j(X),\ \sigma_\epsilon^2 \Big)$$

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The posterior mean of β minimizes

$$\hat{\sigma}^2_{\text{train}} + \frac{1}{N} \sum_j \left(\frac{\sigma_\epsilon}{\tau_j} \right)^2 \beta_j^2,$$

so a prior is a penalty: with all $\tau_j = \tau$ this is ridge regression with $\lambda = \sigma_\epsilon^2 / (N\tau^2)$.

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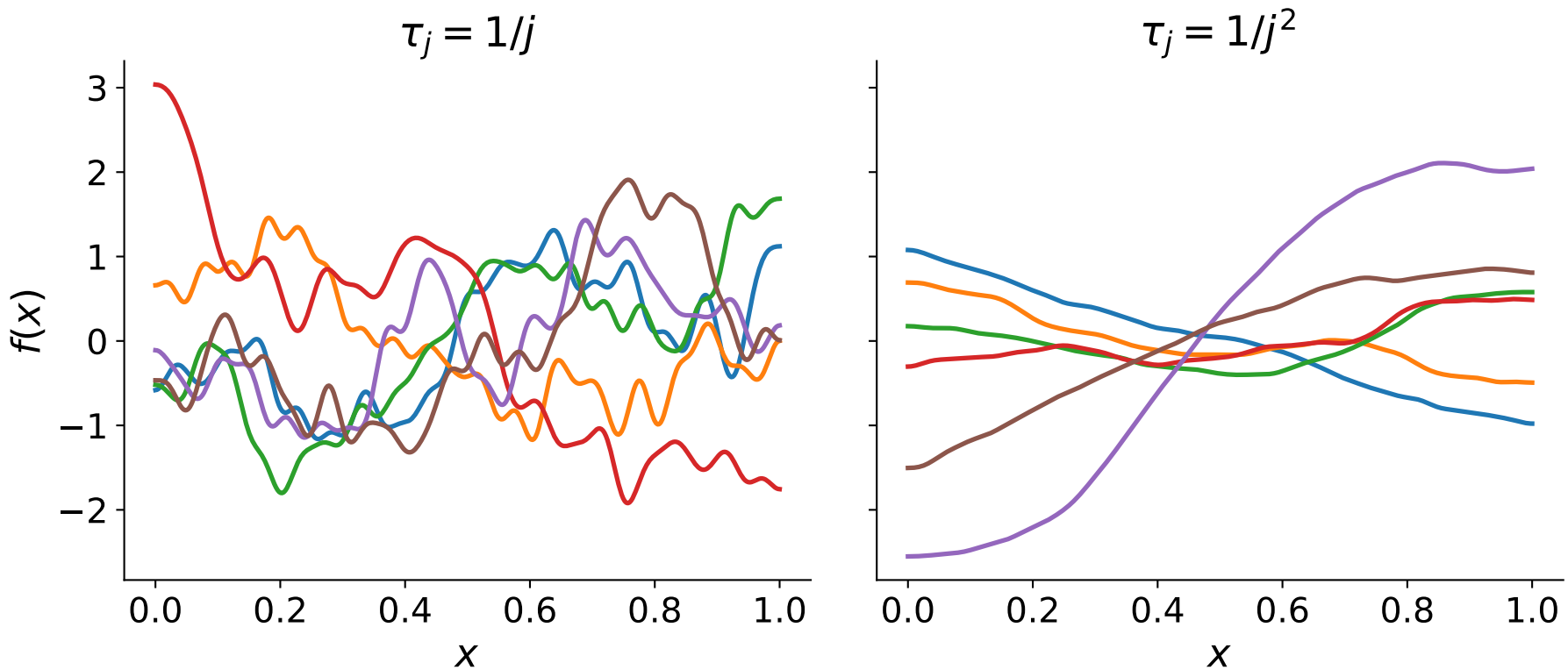
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Posterior distribution of the regression function

Now let's "forget" about the β_j (that is marginalize over them), and examine the distribution of the function $f(x)$. Here are some random samples for different choices of τ_j.



Covariance of a random function

Since $E[\beta_j] = 0$, we have $E[f(x)] = 0$. Since the β_j are independent,

$$\operatorname{cov}\big(f(x), f(x')\big) = \sum_{j=1}^{J} \tau_j^2 \, \phi_j(x)\phi_j(x') = K(x, x').$$

K is called the kernel.

Question

Suppose we observe the value of f at a single point x'. What is $E\big[f(x) \mid f(x')\big]$?

Solution

Both $f(x)$ and $f(x')$ have mean 0, and

$$\text{var}\big(f(x')\big) = K(x', x'), \qquad \text{cov}\big(f(x), f(x')\big) = K(x, x').$$

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Regress $f(x)$ (response) on $f(x')$ (predictor). The intercept is 0 and the slope is

$$\frac{\text{cov}\big(f(x), f(x')\big)}{\text{var}\big(f(x')\big)} = \frac{K(x, x')}{K(x', x')}.$$

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$$\frac{\text{cov}\big(f(x), f(x')\big)}{\text{var}\big(f(x')\big)} = \frac{K(x, x')}{K(x', x')}.$$

Since $f(x)$ and $f(x')$ are sums of Normals, they are jointly Normal, so the regression line is the conditional mean:

$$E\big[f(x) \mid f(x')\big] = \frac{K(x, x')}{K(x', x')}\, f(x').$$